Applied Linear Algebra

Finite-Dimensional Vector Spaces

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Scalars

- A scalar is a single value, which can be an integer, a real number, or a complex number.
- Scalars are the simplest mathematical objects.

Quiz

Name a few good examples of scalars.

Vectors

- A vector is a more structured mathematical object.
- A vector is an ordered sequence of scalars.

$$\mathbf{v} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$$

where $\mathbb{R}^n$ is the set of all n-dimensional real vectors.

$$\mathbf{v} = (0.35, 0.71, 0.23, 0.8) \in \mathbb{R}^4$$

$$\mathbf{v} = (0.35) \in \mathbb{R}^?$$

$$\mathbf{v} = (0.35, 0.71) \in \mathbb{R}^?$$

$$\mathbf{v} = () \in \mathbb{R}^?$$

Let $\mathbf{A}$ be a $m \times n$ matrix.

The n columns of $\mathbf{A}$ are vectors in $\mathbb{R}^m$.

Example: A matrix with 4 column vectors in $\mathbb{R}^5$

$$\begin{bmatrix} .1 & \boxed{\begin{matrix} .2 \\ .5 \\ .8 \\ .12 \\ .22 \end{matrix}} & .3 & 0.6 \\ .4 & & .6 & 0.12 \\ .7 & & .9 & 0.18 \\ .11 & & .13 & 0.26 \\ .21 & & .23 & 0.33 \end{bmatrix}$$

Vectors

A vector can be represented as a row or column of scalars.

Row vector	Column vector
$\begin{pmatrix} 0.35, & 0.71, & 0.23, & 0.8 \end{pmatrix}$	$\begin{bmatrix} 0.35 \\ 0.71 \\ 0.23 \\ 0.8 \end{bmatrix}$

Unit Vector

A *unit vector* is a vector with all elements equal to zero, except one element which is equal to one.

The *i*-th unit vector (of size *n*) is the unit vector with *i*-th element one, and denoted $\mathbf{e}_i$.

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

Unit Vector

The *i*-th unit vector (of size *n*) is the unit vector with *i*-th element one, and denoted $\mathbf{e}_i$.

$$(\mathbf{e}_i)_j = \begin{cases} 1 & i = j \\ 0 & otherwise \end{cases} \tag{1}$$

for $i, j = 1, \ldots, n$.

$\mathbf{e}_i$ is an *n*-vector. $(\mathbf{e}_i)_j$ is a number, its *j*-th entry.

Ones Vector

A *ones vector* is a vector with all elements equal to one. The n-dimensional ones vector is denoted $\mathbf{1}_n$.

$$\mathbf{1}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{1}_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{1}_5 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Properties of Vector Addition

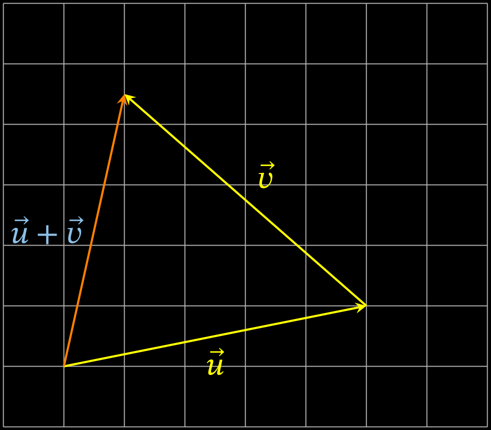
For any n-vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, the following properties hold:

- Vector addition is *commutative*: $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$.
- Vector addition is *associative*: $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$. We can therefore write both as $\mathbf{a} + \mathbf{b} + \mathbf{c}$.
- $\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a}$. Adding the zero vector to a vector has no effect.[†]
- $\mathbf{a} - \mathbf{a} = \mathbf{0}$. Subtracting a vector from itself gives the zero vector.

Can you think of the geometric interpretation of these properties?

[†] The size of the zero vector follows from the context

Vector Addition



Scalar-vector multiplication

For any $\mathbf{a} \in \mathbb{R}^n$, and scalars $\alpha, \beta, \gamma \in \mathbb{R}$, the following properties hold:

- scalar-vector multiplication is *commutative*: $\alpha\mathbf{a} = \mathbf{a}\alpha$.
- scalar-vector multiplication is *associative*: $(\beta\gamma)\mathbf{a} = \beta(\gamma\mathbf{a})$.
- $1\mathbf{a} = \mathbf{a}$. Multiplying a vector by the scalar 1 has no effect.
- $0\mathbf{a} = \mathbf{0}$. Multiplying a vector by the scalar 0 gives the zero vector.
- $\beta(\mathbf{a} + \mathbf{b}) = \beta\mathbf{a} + \beta\mathbf{b}$. Scalar multiplication distributes over vector addition.
- $(\beta + \gamma)\mathbf{a} = \beta\mathbf{a} + \gamma\mathbf{a}$. The distributive property of scalar-vector multiplication.

Vector Spaces

A vector space V is a set endowed with a rule for addition and a rule for scalar multiplication

$$+ : V \times V \to V$$
$$\cdot : \mathbb{R} \times V \to V$$

such that the following axioms hold for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars $c_1, c_2 \in \mathbb{R}$:

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
3. $\mathbf{v} + \mathbf{0} = \mathbf{v}$
4. $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ where $-\mathbf{v}$ is the unique additive inverse of $\mathbf{v}$
5. $c_1 \cdot (\mathbf{v} + \mathbf{w}) = c_1 \cdot \mathbf{v} + c_1 \cdot \mathbf{w}$
6. $(c_1 + c_2) \cdot \mathbf{v} = c_1 \cdot \mathbf{v} + c_2 \cdot \mathbf{v}$
7. $c_1 \cdot (c_2 \cdot \mathbf{v}) = (c_1 c_2) \cdot \mathbf{v}$
8. $1 \cdot \mathbf{v} = \mathbf{v}$

Examples of Vector Spaces

Vector Spaces - Questions

1. is a 3×2 matrix of $\mathbb{R}$ a vector space?
2. is a 3×2 matrix of $\mathbb{R}$ with non-negative entries a vector space?
3. is a $m \times n$ matrix of $\mathbb{R}$ a vector space?

Vector Spaces - Questions

1. Suppose addition in $\mathbb{R}^2$ adds an extra 1 to each component. With scalar multiplication unchanged, which rules are broken?

2. Show that the set of all positive real numbers, with $x + y$ and cx redefined to equal the usual xy and x^c, is a vector space. What is the "zero vector"?

Linear Combinations

if $\mathbf{a}_1, \dots, \mathbf{a}_m$ are n-vectors, and $\beta_1, \dots, \beta_m$ are scalars, the n-vector

$$\beta_1 \mathbf{a}_1 + \cdots + \beta_m \mathbf{a}_m$$

is called the *linear combination* of the vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$.

The scalars $\beta_1, \dots, \beta_m$ are called the *coefficients* of the linear combination.

Linear combination of unit vectors

$$\begin{bmatrix} 8.3 \\ -0.2 \\ 10 \end{bmatrix} = 8.3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-0.2) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 10 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Affine Combination

When the coefficients sum to one, i.e., $\beta_1 + \cdots + \beta_m = 1$, the linear combination is called an *affine combination*.

When the coefficients in an affine combination are nonnegative, it is called a *convex combination*, or a *weighted average*.

Inner Product

Definition (Inner or Dot Product)

The inner product of two n-vectors is defined as the *scalar*

$$\mathbf{a}^{\mathrm{T}}\mathbf{b} = a_1 b_1 + a_2 b_2 + \ldots + a_n b_n,$$

the sum of the products of corresponding entries.

If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors of the same size, and γ is a scalar, then the following hold:

1. *Commutativity*: $\mathbf{a}^{\mathrm{T}}\mathbf{b} = \mathbf{b}^{\mathrm{T}}\mathbf{a}$ aka *symmetric*
2. *Associativity with scalar multiplication*: $(\gamma\mathbf{a})^{\mathrm{T}}\mathbf{b} = \gamma(\mathbf{a}^{\mathrm{T}}\mathbf{b})$
3. *Distributivity with vector addition*: $(\mathbf{a}+\mathbf{b})^{\mathrm{T}}\mathbf{c} = \mathbf{a}^{\mathrm{T}}\mathbf{c} + \mathbf{b}^{\mathrm{T}}\mathbf{c}$

Inner Product: Special Examples

- *Unit vector.* $e_i^{\mathrm{T}}\mathbf{a} = a_i$
- *Sum.* $1^{\mathrm{T}}\mathbf{a} = a_1 + \cdots + a_n$
- *Sum of squares.* $\mathbf{a}^{\mathrm{T}}\mathbf{a} = a_1^2 + \cdots + a_n^2$

Inner Product: Applications

- *Co-occurrence.* $\mathbf{a}^{\mathrm{T}}\mathbf{b} = a_1 b_1 + \cdots + a_n b_n$ counts the number of times the two vectors have non-zero entries in the same position.
- *Weighs, features, and scores.* $\mathbf{w}^{\mathrm{T}}\mathbf{x} = w_1 x_1 + \cdots + w_n x_n$
- *Polynomial evaluation.* $p(x) = a_0 + a_1 x + \cdots + a_n x^n = [1, x, \ldots, x^n]^{\mathrm{T}}[a_0, a_1, \ldots, a_n]$
- *Probability and expected values.* $E[X] = \sum_{i=1}^{n} x_i p_i = [x_1, \ldots, x_n]^{\mathrm{T}}[p_1, \ldots, p_n]$

Outer Product

Definition (Outer Product)

The outer product of a *m*-vector **a** and an *n*-vector **b** is given by $\mathbf{a}\mathbf{b}^{\mathrm{T}}$, which is an m×n matrix

$$\mathbf{a}\mathbf{b}^{\mathrm{T}} = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & a_2 b_2 & \cdots & a_2 b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_m b_1 & a_m b_2 & \cdots & a_m b_n \end{bmatrix},$$

whose entries are all products of the entries of a and the entries of b.

In general, $\mathbf{a}^{\mathrm{T}}\,\mathbf{b} \neq \mathbf{b}^{\mathrm{T}}\,\mathbf{a}$, *i.e.*, it is not *symmetric*

The Zero-Dimensional Vector Space $\mathbb{R}^0$

Definition (The Trivial Space)

The space $\mathbb{R}^0$ is the unique zero-dimensional real vector space. It contains exactly one element, the empty vector $\mathbf{v}$:

$$\mathbf{v} \in \mathbb{R}^0 \quad \text{where} \quad \mathbf{v} = []$$

Subspaces

Definition
A subspace W of a vector space V is a non-empty subset of V that is itself a vector space under the same operations of addition and scalar multiplication defined on V.

1. The zero vector of V is in W.
2. If $u, v \in W$, then $u + v \in W$ (closed under addition).
3. If c is a scalar and $v \in W$, then $cv \in W$ (closed under scalar multiplication).

Vector subspaces

Consider all vectors in $\mathbb{R}^2$ whose components are positive or zero.

This subset is the first quadrant of the x-y plane.

It is not a subspace of $\mathbb{R}^2$ because it is not closed under scalar multiplication.

For example, if

$$\mathbf{v} = [1, 1]$$

is in the first quadrant, then

$$-1 \cdot \mathbf{v} = [-1, -1]$$

is not in the first quadrant.

Subspaces

If we include the third quadrant along with the first, scalar multiplication is closed, but the set is still not a subspace because it is not closed under addition.

For example, if

$$\mathbf{u} = [1, 2] \quad \text{and} \quad \mathbf{v} = [-2, -1]$$

are in the first and third quadrants, then

$$\mathbf{u} + \mathbf{v} = [-1, 1]$$

is not in the first or third quadrant.

The smallest subspace of $\mathbb{R}^2$ that contains the first quadrant is the entire space $\mathbb{R}^2$.

Math vs. Code

— Textbooks use *column vectors* for single samples.

— PyTorch uses *row vectors* to handle batches.

- Single sample $\mathbf{x}$ is a column vector.
- Forward pass formula:

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$$

- Dimensions:

$$\underset{(c\times 1)}{\mathbf{y}} = \underset{(c\times d)}{\mathbf{W}} \; \underset{(d\times 1)}{\mathbf{x}} + \underset{(c\times 1)}{\mathbf{b}}$$

Math vs. Code

— Textbooks use *column vectors* for single samples.

— PyTorch uses *row vectors* to handle batches.

- Batch $\mathbf{X}$ stacks samples as rows.
- PyTorch *nn.Linear* formula:

$$\mathbf{Y} = \mathbf{X}\mathbf{W}^T + \mathbf{B}$$

- Dimensions (N = batch size):

$$\underset{(N\times c)}{\mathbf{Y}} = \underset{(N\times d)}{\mathbf{X}} \; \underset{(d\times c)}{\mathbf{W}^T} + \underset{(1\times c)}{\mathbf{B}}$$

NumPy and PyTorch

The standard convention for data analysis and machine learning is that

- rows represent samples or observations: **axis 0**
- columns represent features or attributes: **axis 1**